

Trapezoidal Grid Reconstruction for Efficient Non-Line-of-Sight Imaging

Supplementary Material

In Section S1, we provide the mathematical derivations referenced in the main paper. Section S2 details the hardware setup for NLOS imaging used to collect the experimental datasets. In Section S3, we present additional results.

S1 DERIVATIONS

S1.1 Scaled FFT Derivation

Starting with the DFT equation:

$$U[m', n'] \triangleq \sum_{m=-M/2}^{M/2-1} \sum_{n=-N/2}^{N/2-1} u[m, n] \cdot k_M^{-m \cdot m'} k_N^{-n \cdot n'} \quad (S1)$$

where

$$k_L = \exp \left[-j \frac{2\pi}{L} \right] \quad (S2)$$

For the rest of the derivation, we use (m, n) to represent the **integer** coordinates in the input space with $\delta_{\text{in}} = \delta$ sampling interval, and (m', n') to represent the **integer** coordinates in the output space with $\delta_{\text{in}} = \delta$ sampling rate. Note that (m, n, m', n') are all variables that range from $-N/2$ to $N/2 - 1$.

For a given frequency, we simplify the input phasor field $\mathcal{P}_{\mathcal{F}}(x_c, y_c, 0, \omega)$ to a discrete 2D representation $u[m, n]$, and the goal is to compute its Fourier transform on a scaled grid, i.e., $U[m', n']$. Use the definition of DFT, evaluating the output at the scaled coordinates introduces an extra α factor in the exponent term:

$$\begin{aligned} U_{\alpha}[m', n'] &\triangleq \mathcal{F} \{ \mathcal{P}_{\mathcal{F}}(\alpha x_c, \alpha y_c, 0, \omega) \} \\ &= \sum_{m=-N/2}^{N/2-1} \sum_{n=-N/2}^{N/2-1} u[m, n] \cdot \exp \left[\left(-j \frac{2\pi}{N} \right) \cdot -\alpha (m \cdot m' + n \cdot n') \right]. \end{aligned} \quad (S3)$$

$U_{\alpha}[m', n']$ is essentially a squeezed and resampled version of $U[m', n']$, but we stay with the integer indices notation using m' and n' because all the signals are supposed to be discrete. In the derivation that follows, we demonstrate that explicit zero-padding is not required to achieve the desired resampling of $U[m', n']$. Instead, the process can be reformulated as a set of additional FFTs, thereby accelerating the computation. We here start from the 1D case and then extend into 2D later.

$$U_{\alpha}[n'] = \sum_{n=-N/2}^{N/2-1} u[n] \cdot k_N^{-\alpha \cdot n \cdot n'} \quad \text{where} \quad k_N = \exp \left[-j \frac{2\pi}{N} \right] \quad (S4)$$

since $\alpha \cdot n \cdot n' = \alpha/2 \cdot (n^2 + n'^2 - (n' - n)^2)$, Eq. S3 can be rewritten as:

$$\begin{aligned} U_{\alpha}[n'] &= \sum_{n=-N/2}^{N/2-1} u[n] \cdot k_N^{\frac{\alpha}{2}(n'-n)^2} \cdot k_N^{-\frac{\alpha}{2}n^2} \cdot k_N^{-\frac{\alpha}{2}n'^2} \\ &= k_N^{-\frac{\alpha}{2}n'^2} \sum_{n=-N/2}^{N/2-1} \underbrace{u[n] \cdot k_N^{-\frac{\alpha}{2}n^2}}_{u'[n]} \cdot \underbrace{k_N^{\frac{\alpha}{2}(n'-n)^2}}_{k'_N[n'-n]} \\ &= k_N^{-\frac{\alpha}{2}n'^2} \sum_{n=-N/2}^{N/2-1} u'[n] \cdot k'_N[n'-n] \end{aligned} \quad (S5)$$

After defining $u'[n] = u[n] \cdot k_N^{-\frac{\alpha}{2}n^2}$ and $k'_N[n] = k_N^{\frac{\alpha}{2}n^2}$, Eq. S5 can be treated as a convolution and apply FFT to calculate.

$$U_{\alpha}[n'] = k_N^{-\frac{\alpha}{2}n'^2} \cdot \text{IFFT} \{ \text{FFT} \{ u'[n] \} \cdot \text{FFT} \{ k'_N[n] \} \} \quad (S6)$$

S1.2 Volume of Frustum

For a pyramidal frustum, the side length varies linearly with z so that:

$$x_{\text{out}}(z) = x_{\text{in}} + k_x z, \quad \alpha(z) = \frac{x_{\text{in}}}{x_{\text{in}} + k_x z} \quad (S7)$$

$$y_{\text{out}}(z) = y_{\text{in}} + k_y z, \quad \beta(z) = \frac{y_{\text{in}}}{y_{\text{in}} + k_y z} \quad (S8)$$

$$z_{\text{out}} = z_{\text{in}} + z \quad (S9)$$

Then we can derive the volume by integrating cross-sectional areas along z :

$$\begin{aligned}
V_F &= \int_{z_{\text{in}}}^{z_{\text{out}}} A(z) dz \\
&= \int_{z_{\text{in}}}^{z_{\text{out}}} (x_{\text{in}} + \alpha z) (y_{\text{in}} + k_y z) dz \\
&= \int_{z_{\text{in}}}^{z_{\text{out}}} (x_{\text{in}} y_{\text{in}} + k_y x_{\text{in}} z + k_x y_{\text{in}} z + (k_x k_y z^2)) dz \\
&= \left| \left(z (x_{\text{in}} y_{\text{in}}) + \frac{z^2}{2} (k_y x_{\text{in}} + k_x y_{\text{in}}) + \frac{z^3}{3} (k_x k_y) \right) \right|_{z_{\text{in}}}^{z_{\text{out}}}
\end{aligned} \tag{S10}$$

Assuming that each cross-section of the frustum is a square, then $x_{\text{in}} = y_{\text{in}}$ and $k_x = k_y = k$, we get:

$$V_F = \left| \left(z x_{\text{in}}^2 + z^2 x_{\text{in}} k + z^3 \left(\frac{k^2}{3} \right) \right) \right|_{z_{\text{in}}}^{z_{\text{out}}} \tag{S11}$$

Letting $z_{\text{in}} = 0$, $z_{\text{out}} = H$, we recover Eq. 24:

$$V_F = \left(H x_{\text{in}}^2 + H^2 x_{\text{in}} k + H^3 \left(\frac{k^2}{3} \right) \right).$$

where H is the height of our frustum.

S1.3 Resolution-Safe Sampling Bound

Under isotropic scaling, the lateral field of view increases with depth as

$$x_{\text{out}}(z) = x_{\text{in}} + k z, \tag{S12}$$

which implies a depth-dependent lateral voxel spacing

$$\delta_{\text{out}}(z) = \delta_{\text{in}} \left(1 + \frac{k z}{x_{\text{in}}} \right). \tag{S13}$$

To avoid under-sampling resolvable structure, the output voxel spacing must satisfy

$$\delta_{\text{out}}(z) \leq \frac{\Delta x(z)}{2}, \tag{S14}$$

where $\Delta x(z)$ is the depth-dependent lateral resolution. Using the diffraction-limited form

$$\Delta x(z) = \frac{1.22 \lambda_c z}{D}, \tag{S15}$$

we define the corresponding resolution limit

$$\delta_{\text{limit}}(z) \triangleq \frac{\Delta x(z)}{2}. \tag{S16}$$

Enforcing (S14) at the farthest depth $z = z_{\text{max}}$ gives

$$\delta_{\text{in}} \left(1 + \frac{k z_{\text{max}}}{x_{\text{in}}} \right) \leq \delta_{\text{limit}}(z_{\text{max}}), \tag{S17}$$

which yields the resolution-safe upper bound

$$k \leq k_{\text{max}} = \frac{x_{\text{in}}}{z_{\text{max}}} \left(\frac{\delta_{\text{limit}}(z_{\text{max}})}{\delta_{\text{in}}} - 1 \right). \tag{S18}$$

This condition can be interpreted as a depth-dependent Nyquist criterion for the scaled voxel grid. In the experiments below, we report the normalized ratio $\delta_{\text{out}}/\delta_{\text{limit}}$ to indicate whether the voxel spacing remains within the resolution-safe range.

S2 HARDWARE

In this work, we build an active NLOS imaging system with a PM-1.03-25TM pulsed laser from Polar Laser Laboratories as the illumination source, and a customized gated 16 x 16 pixel Single Photon Avalanche Diode (SPAD) array, developed at Politecnico di Milano [36], as the detector (Column 1, Fig. S1). The laser is coupled with a frequency doubler to generate 515 nm pulses at an average rep rate of 5 MHz and average power of 375 mW. We sequentially illuminate individual locations on a 1.9 m x 1.9 m area on a relay wall using a two-mirror Thorlabs galvanometer (Thorlabs GVS012). Furthermore, we use an Edmund Optics 6X Manual Zoom Video lens to focus our 16x16 array to a 50 cm x 32.5 cm (Width x Height) area on the relay wall. The overall temporal resolution of our system is characterized by a Full-Width at Half Maximum (FWHM) of approximately 65 ps.

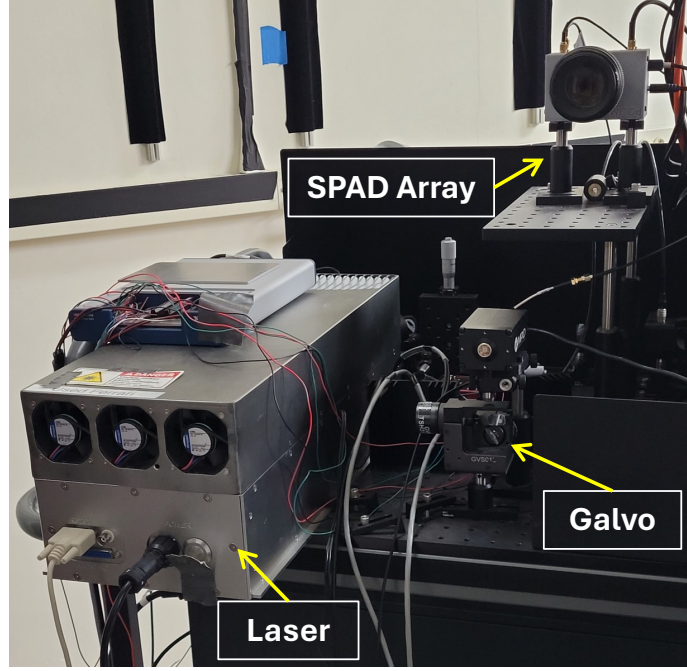


Fig. S1: An image of our NLOS imaging system consisting of a pulsed laser, galvanometer, and SPAD array.

S3 SUPPLEMENTAL RESULTS

S3.1 Empirical Validation of the Resolution-Safe Bound

Figure S2 shows reconstructions obtained for increasing values of the normalized sampling ratio $\delta_{\text{out}}/\delta_{\text{limit}}$, where δ_{limit} is defined in (S16) and the corresponding resolution-safe condition is given by (S14). All other parameters are held fixed. For ratios below unity, the reconstructions remain stable and preserve the hidden object structure. As the ratio approaches unity, mild degradation becomes visible. Once the bound is exceeded, ringing and structured aliasing artifacts appear and rapidly dominate the reconstruction. This behavior is consistent with Fourier-domain aliasing: when the scaled voxel spacing exceeds the resolution-safe limit, resolvable spatial frequencies are no longer sampled adequately and fold back into the reconstructed image.

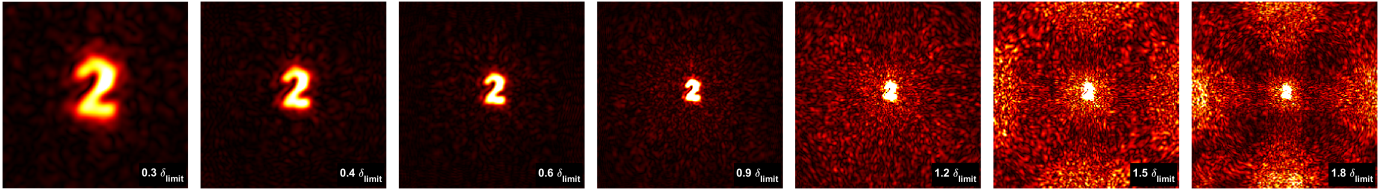


Fig. S2: Effect of exceeding the resolution-safe sampling limit. Reconstructions for increasing values of the normalized ratio $\delta_{\text{out}}/\delta_{\text{limit}}$, with all other parameters fixed. Below the limit, the hidden object remains stable and interpretable. Near the limit, mild degradation appears. Beyond the limit, ringing and structured aliasing artifacts rapidly degrade the reconstruction.

S3.2 Scaled RSD on Non-Uniform Acquisition schemes

The Scaled RSD algorithm requires the measurement to be collected with a uniformly sampled spatial grid on the relay surface, which typically requires additional calibration. One strategy that has worked for other RSD based methods [37], is to sample the relay surface with a non-uniform grid, indicated by $\vec{x}'_c$, and interpolate to a regular grid ($\vec{x}_c$):

$$\mathcal{P}_{\mathcal{F}}(\vec{x}_c, \omega) = \phi(\mathcal{P}_{\mathcal{F}}(\vec{x}'_c, \omega)) \quad (\text{S19})$$

Then, we can proceed with the Scaled RSD as usual. In Fig. S3, we demonstrate that this strategy works well on experimental data from [19]. This dataset was collected with a uniform grid — to recreate a non-uniform acquisition grid, we randomly subsample the relay surface while simultaneously discarding 90% of the grid positions (Column 1). Then we reconstruct with the total grid (Row 1) and the non-uniform sparse grid (Row 2) using the Standard RSD (Columns 2 and 4) and the Scaled RSD (Columns 3 and 5). Our results show that the Scaled RSD can generate reconstructions using non-uniform sparse grids without substantial loss in image quality.

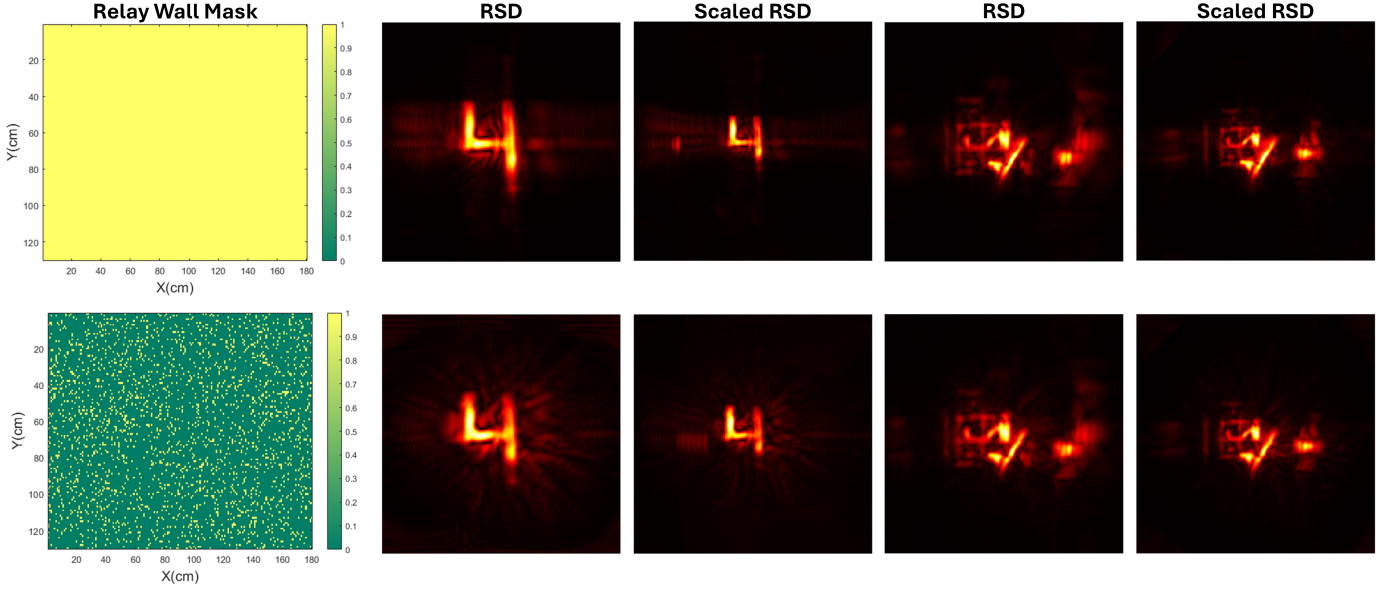


Fig. S3: In Row 1, we show the reconstructions for two different datasets using the Standard RSD (Columns 2 and 4) and Scaled RSD (Columns 3 and 5). In Row 2, we randomly subsample 10% of the relay surface (Column 1) for the same datasets and interpolate to a uniform grid. We show that both Standard RSD (Columns 2 and 4) and the Scaled RSD (Columns 3 and 5) reconstructions do not degrade in quality after discarding 90% of grid positions. Scaled RSD generates reconstructions with perspective projection.

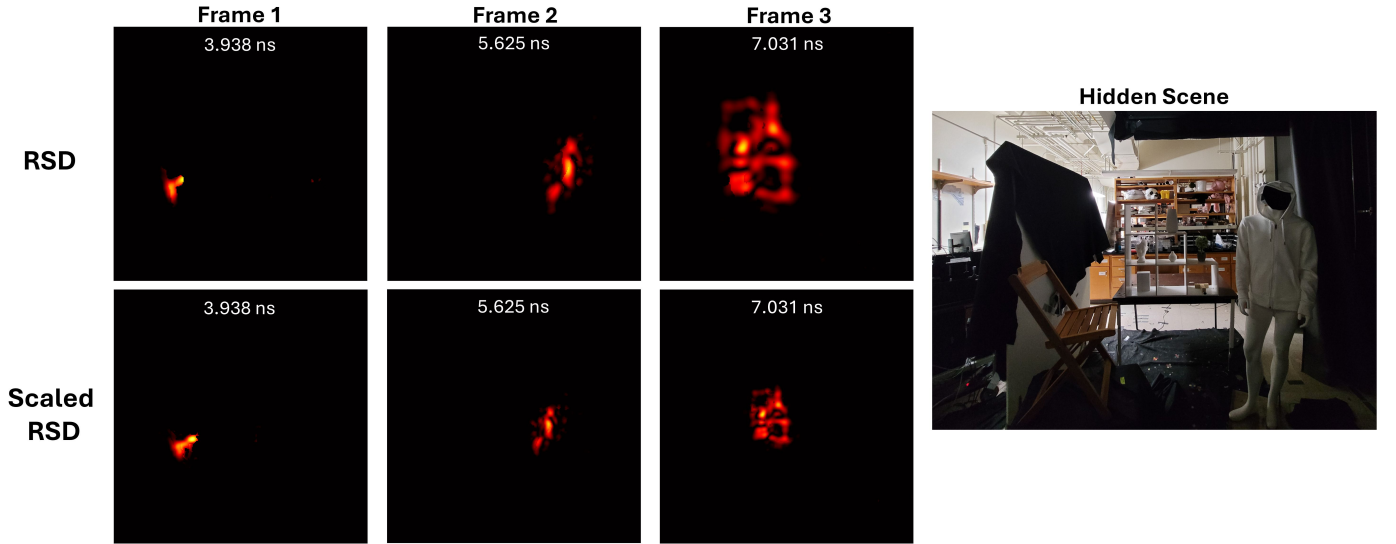


Fig. S4: We generate 4D videos of light transport in the hidden scene using the RSD (Row 1) and the Scaled RSD (Row 2). The mannequin and shelf towards the back of the hidden volume appear smaller in the scaled RSD (Row 2, Columns 3 and 4) due to perspective projection

S3.3 4D Light Transport

The Standard RSD algorithm can be used to generate a 4D video of our virtual source propagating and interacting with the hidden scene [9]. This algorithm can be adapted to extract complex light transport of the hidden scene [38].

$$I(\vec{x}_v, t) = \iiint \mathcal{P}_{\mathcal{F}}(\vec{x}_p, \omega) e^{-j\frac{\omega}{c}(|\vec{x}_c - \vec{x}_v|)} e^{j\omega t} d\omega d\vec{x}_c d\vec{x}_p \quad (\text{S20})$$

In Row 2 of Fig. S4, we show that the Scaled RSD can be used to generate these 4D videos of the hidden scene, while maintaining perspective projection.

S3.4 Depth-resolved comparison to spherical-grid FBP

To further quantify the relationship between spherical sampling, trapezoidal sampling, and Scaled RSD, we compare reconstructions across the depth sequence shown in Fig. 5. For each depth, spherical-grid FBP is used as the reference. We evaluate two methods against this reference: direct FBP on a trapezoidal grid and Scaled RSD on the same trapezoidal grid. All reconstructions

TABLE S1: Depth-resolved SSIM comparison using spherical-grid FBP as the reference. All methods are evaluated on maximum-intensity projections after maximum-intensity normalization, orientation matching, and 2D correlation alignment. Masked SSIM is computed over pixels above 10% of the spherical FBP reference peak intensity.

Depth (m)	FBP + Trapezoidal		Scaled RSD + Trapezoidal	
	SSIM	Masked SSIM	SSIM	Masked SSIM
1.20	0.913	0.987	0.879	0.977
1.48	0.947	0.995	0.929	0.992
1.80	0.929	0.996	0.927	0.992
2.13	0.929	0.994	0.939	0.995
2.84	0.935	0.996	0.896	0.996
3.32	0.940	0.996	0.849	0.992
3.65	0.913	0.995	0.825	0.990

TABLE S2: Depth-resolved PSNR comparison using spherical-grid FBP as the reference. PSNR is reported in dB. Masked PSNR is computed over pixels above 10% of the spherical FBP reference peak intensity.

Depth (m)	FBP + Trapezoidal		Scaled RSD + Trapezoidal	
	PSNR	Masked PSNR	PSNR	Masked PSNR
1.20	33.02	34.54	30.43	31.61
1.48	34.40	35.46	33.81	35.17
1.80	34.94	37.01	34.39	35.91
2.13	33.92	35.58	34.18	35.61
2.84	34.78	36.32	34.14	37.98
3.32	36.69	39.21	32.30	37.00
3.65	35.79	38.71	31.52	37.46

are converted to 2D maximum-intensity projections, normalized by their maximum intensity, orientation matched, and aligned using 2D correlation before computing SSIM and PSNR.

The trapezoidal-grid FBP reconstructions remain close to the spherical-grid FBP reference across all depths, with SSIM values between 0.913 and 0.947. This supports the use of the trapezoidal grid as a practical approximation to spherical sampling. Scaled RSD also shows strong agreement with the spherical reference at near and intermediate depths, with SSIM values above 0.92 for several depths. At larger depths, the global SSIM decreases, reaching 0.825 at 3.65 m. This decrease is primarily due to background and artifact differences becoming more influential as the projected object occupies fewer pixels. Consistent with this interpretation, masked SSIM computed over the object-support region remains high across all depths, staying above 0.977 for Scaled RSD.

These results show that Scaled RSD preserves the main object structure of the corresponding trapezoidal/spherical reconstructions while avoiding direct spherical-grid evaluation. The comparison also reinforces the intended scope of Scaled RSD: it improves depth-adaptive voxel allocation, memory use, and runtime efficiency, rather than increasing the intrinsic far-range resolution of the NLOS measurement.

S3.5 CPU-Time Scaling under Matched Coverage

We empirically evaluate computational scaling by varying the maximum reconstruction depth from 1.4 m to 10 m while holding the minimum depth at 1.0 m, the depth spacing at 0.04 m, and the near-plane side length at 1.28 m. For each maximum depth, all methods reconstruct matched depth-scaled lateral coverage. Scaled RSD retains 128×128 lateral samples per depth plane, whereas Standard RSD increases the size of its enclosing Cartesian grid to cover the same expanding field of view. At shallow depths, the additional FFT operations make Scaled RSD comparable to or slightly slower than Standard RSD. As the reconstruction depth increases, however, the cost of the enlarged Cartesian grid grows substantially faster. At $z_{\max} = 10$ m, Scaled RSD requires 84.3 s, compared with 1490.6 s for Standard RSD, corresponding to approximately a $17.7\times$ reduction in reported CPU time. Direct depth-scaled FBP is substantially more expensive over the evaluated range and was not evaluated for the larger volumes due to its computational cost. These measurements are consistent with the complexity analysis in Sec. 4.3.

S3.6 Cartesian RSD with Spherical Resampling

We evaluate Cartesian RSD followed by spherical resampling as a post-processing baseline for generating a depth-scaled output. Standard RSD first reconstructs a matched-coverage Cartesian volume containing $350 \times 350 \times 71$ voxels, which is then resampled onto a $190 \times 190 \times 71$ spherical grid. Direct spherical-grid FBP with the same output dimensions is used as the reference. Using the full reconstructed images without foreground masking, Cartesian RSD followed by spherical resampling achieves an SSIM of 0.8573 and a PSNR of 29.84 dB relative to spherical FBP. The comparison shows that post-hoc spherical resampling can approximate the direct spherical reconstruction, but requires first computing and storing the larger Cartesian volume.

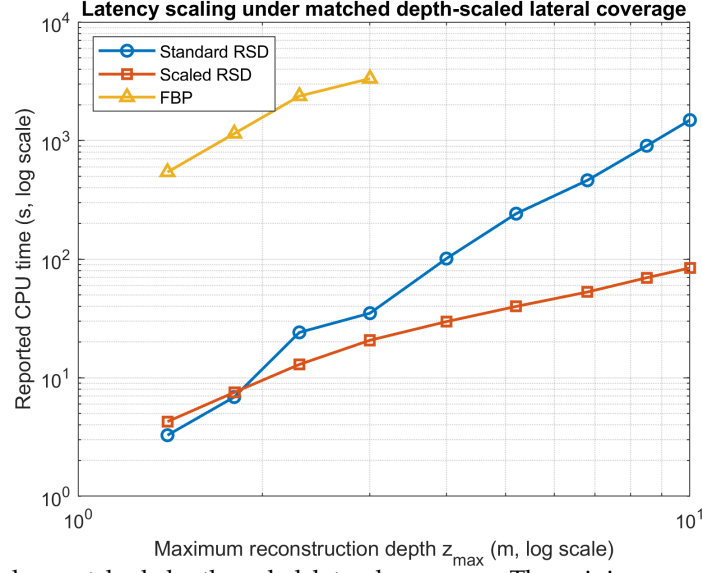


Fig. S5: CPU-time scaling under matched depth-scaled lateral coverage. The minimum reconstruction depth is fixed at 1.0 m, the depth spacing is 0.04 m, and the near-plane side length is 1.28 m. Scaled RSD and depth-scaled FBP use 128×128 lateral samples per depth plane, whereas Standard RSD enlarges its enclosing Cartesian grid to cover the same depth-dependent lateral field of view. All measurements are CPU-only and were obtained on the same workstation. For FBP, we report thread-normalized CPU time using 32 threads. FBP was evaluated only up to $z_{\max} = 3.0$ m because of its substantially higher computational cost. Both axes use logarithmic scales.

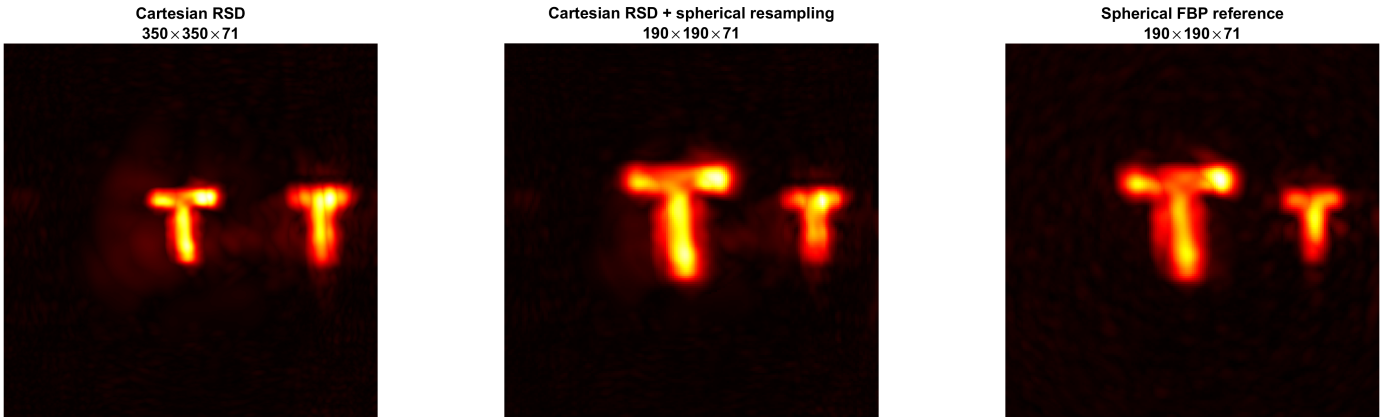


Fig. S6: Comparison of Cartesian RSD, Cartesian RSD followed by spherical resampling, and direct spherical-grid FBP. Standard RSD reconstructs a $350 \times 350 \times 71$ matched-coverage Cartesian volume, which is resampled onto a $190 \times 190 \times 71$ spherical grid. Direct spherical-grid FBP at the same output dimensions is used as the reference. The full-image, unmasked comparison between Cartesian RSD with spherical resampling and spherical FBP gives an SSIM of 0.8573 and a PSNR of 29.84 dB.